

Introduction to Classification

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Lecture 01.2 (v4.0.0)



Chihuahua or muffin?

Signposting

- ▶ Most of machine learning is modern statistics.
 - ▶ The main distinction is about purpose: whether developed for prediction, or for understanding.
- ▶ Today is about the framing of the problem.

Types of machine learning

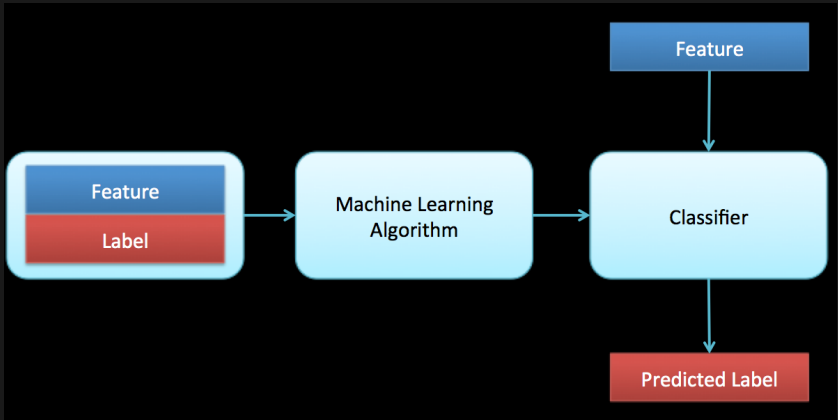
- ▶ Unsupervised: **no labels**. For example,
 - ▶ Clustering
 - ▶ Dimensionality reduction
 - ▶ Smoothing
- ▶ Supervised: exploits **labels**. For example,
 - ▶ Classification
 - ▶ Regression
- ▶ Other types:
 - ▶ Semi-supervised: **some labels** are available
 - ▶ Active: can **choose which labels** to obtain
 - ▶ Reinforcement: **reward based**. explore vs exploit?
 - ▶ etc.

Magic of Supervised Learning

- ▶ By **cross-validation**, splitting data into:
 - ▶ **Train**: What we learn parameters on,
 - ▶ **Calibration**: What we learn hyperparameters on,
 - ▶ **Test**: What we report performance on,
- ▶ Machine Learning can be (almost?) automated -
 - ▶ If you can define a suitable **label** for supervision,
 - ▶ And if you can find **test data** that look like your real problem.

Classification

- ▶ Machine Learning classification is about how to make good predictions of **classes** based on previous experience of how **features** relate to **classes**.

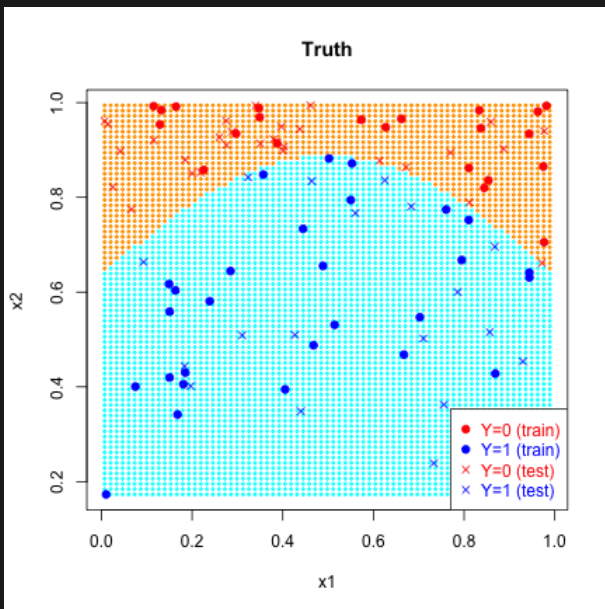


Examples of classification

Classification is broadly the “detection, recognition, recall of **prior experience**”.

- ▶ **Spam** filtering (spam/not spam)
- ▶ **Face detection** (image classification)
- ▶ **Speech recognition**
- ▶ **Handwriting recognition**
- ▶ **Turing test** . . . though that is human, not machine!
- ▶ **Next token** prediction for LLMs. . .

Classification



From Regression to Classification

- ▶ **Linear regression** is obtaining solutions to:

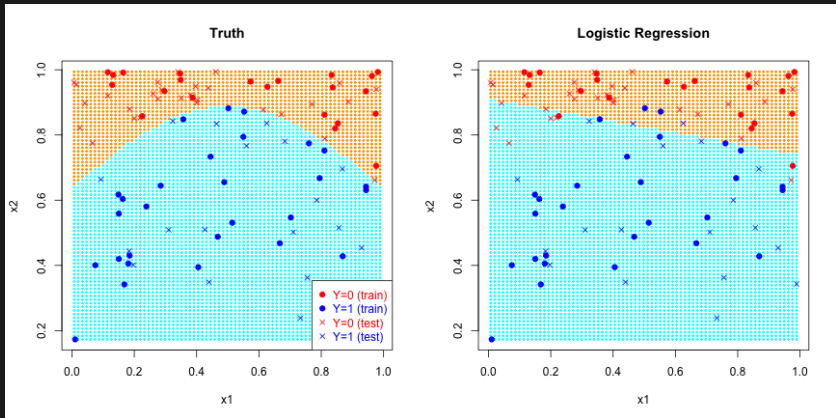
$$y_i = \vec{x}_i \cdot \beta + e_i$$

- ▶ Definitions: $\vec{x}_i$ and β are vectors of length p ; e_i are residuals whose squared-sum is minimised.
- ▶ **Logistic regression** estimates the probability that a binary outcome y is 1:

$$\text{logit}(p(y_i)) = \ln \left(\frac{p(y_i)}{1 - p(y_i)} \right) = \vec{x}_i \cdot \beta + e_i$$

- ▶ with $y_i \sim \text{Bern}(p(y_i))$. The prediction is the **log-odds** ratio, with values > 0 predicting a 1 and values < 0 predicting a 0.

Logistic Regression example



Classification Performance (naive)

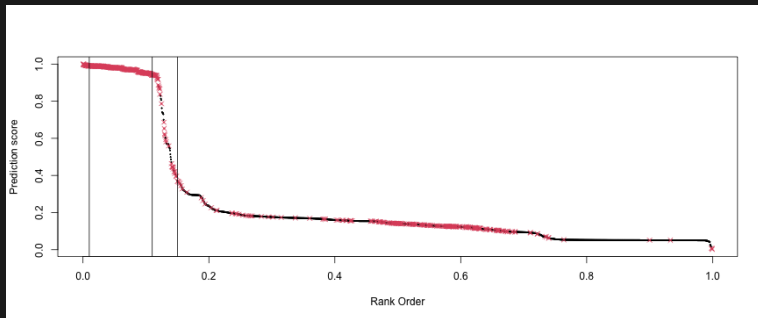
Consider a **best guess** classifier, i.e. we output a predicted class only.

- ▶ Evaluate performance on **test** data, to prevent over-fitting.
- ▶ **Accuracy** is the proportion of correctly classified points.
 - ▶ But... what is the **baseline**? If there are imbalanced classes?
- ▶ Classifier performance is understood through a **Confusion Matrix**, a comparison between:
 - ▶ Ground truth class, and
 - ▶ Predicted classes.
- ▶ For binary classes, summarise using (true/false)(positive/negative) outcomes.

Classification Performance

- ▶ Most predictors are continuous scores, allowing **ranking**
 - ▶ e.g. Logistic regression log-odds
 - ▶ But... are these **probabilities**?
- ▶ Scores are **ordered**. So we can choose a threshold to control the total proportion of **positive predictions**.
 - ▶ This provides a **relationship** between **Positive Claims** and **True Positives**.
 - ▶ Multiclass classification contains nuance

Classification Performance Thresholding

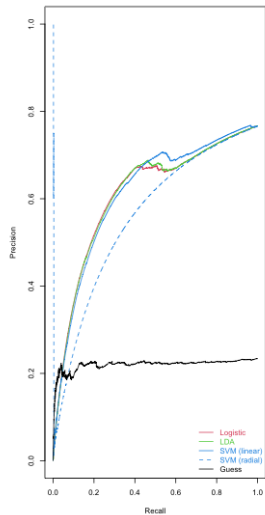
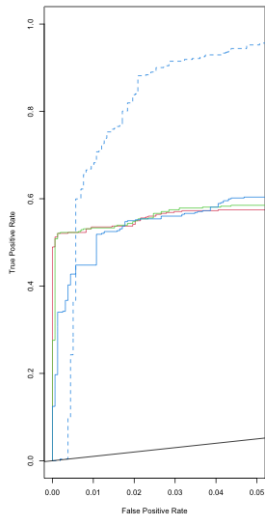
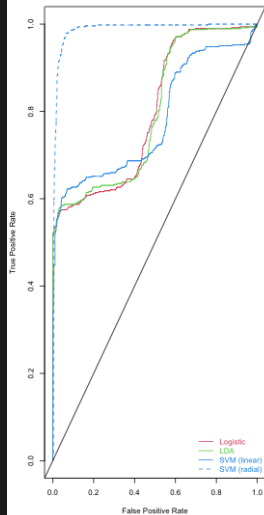


	$Y = 1$	$Y = 0$	Condition
$\hat{Y} = 1$	TP	FP	Prediction positive
$\hat{Y} = 0$	FN	TN	Prediction negative
Claim	Truth positive	Truth negative	

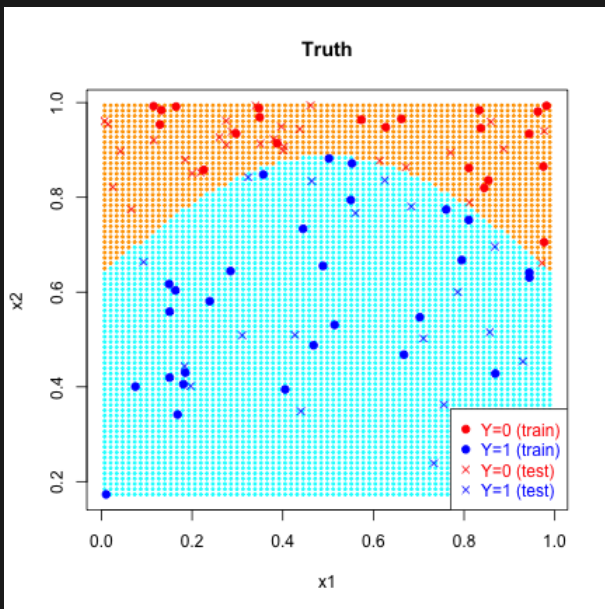
Classification Performance Representations

- ▶ There are many ways to represent performance
- ▶ The **Receiver-Operator-Curve (ROC)** is the most popular, as it holds regardless of the true distribution of the data.
 - ▶ X-axis: False Positive Rate (FPR) = $P(\hat{Y} = 1|Y = 0)$
 - ▶ Y-axis: True Positive Rate (TPR) = $P(\hat{Y} = 1|Y = 1)$
 - ▶ The **Area Under the Curve (AUC)** is a measure of Accuracy (0.5=guessing, 1=perfect).
 - ▶ We need to care about the region of the ROC curve that matters.
- ▶ The **Precision-Recall curve** is appropriate when we care specifically about positive cases:
 - ▶ X-axis: Precision = $P(Y = 1|\hat{Y} = 1)$
 - ▶ Y-axis: Recall=TPR = $P(\hat{Y} = 1|Y = 1)$

ROC/PR Curve Example



Classification



K-Nearest Neighbour classification

- ▶ We later introduce **K-NN**.

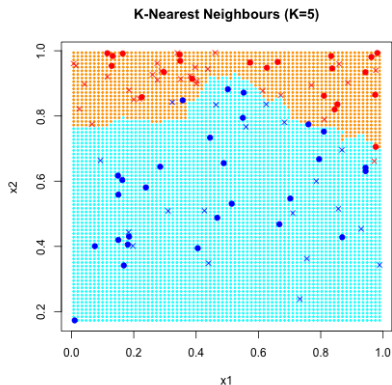
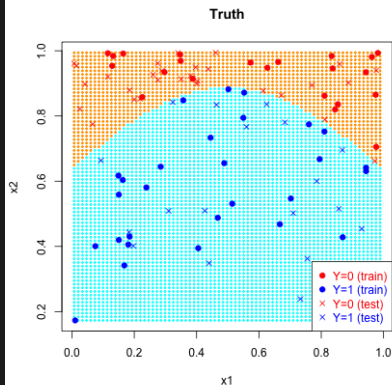
- ▶ Data in $\mathcal{R}^d$
- ▶ Make a choice of distance function, e.g. Euclidean

$$d(x, y) = \|x - y\|_2$$

- ▶ Obtained the K nearest neighbours of points
- ▶ Classifier: **majority vote** of the training labels of all k neighbours.
 - ▶ But... How do you choose k ?
- ▶ A naive implementation scales poorly with N , but an approximate lookup can control complexity.
 - ▶ See also: Condensed nearest neighbor¹ approaches to reduce the amount of data required at the classification stage.

¹Hart P, The Condensed Nearest Neighbor Rule. IEEE Transactions on Information Theory 18 (1968) 515-516. doi: 10.1109/TIT.1968.1054155

K-Nearest Neighbour example



Wrapup

- ▶ These are **baseline models**
- ▶ **Logistic regression** is the go-to straw man classifier in machine learning:
 - ▶ It is a natural predictive model
 - ▶ It does surprisingly well in many settings
- ▶ **k-NN** is the interpolation method to beat
 - ▶ **Neighbourhoods** are always competitive, but are costly at test time
 - ▶ How should we measure “close”?
- ▶ The magic is in the **sampling**:
 - ▶ train, calibration, **test**

Portfolio:

- ▶ Portfolio Question 1.2.P: What metrics are better than accuracy? In what sense are they reliable, and do they relate to uncertainty?

References:

- ▶ **References** for classification basics:
 - ▶ Stack Exchange Discussion of ROC vs PR curves.
 - ▶ Davis and Goadrich, "The Relationship Between Precision-Recall and ROC Curves", ICML 2006.
 - ▶ Rob Schapire's ML Classification features a Batman Example...
 - ▶ Chapter 4 of *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (Friedman, Hastie and Tibshirani).
- ▶ k-Nearest Neighbours:
 - ▶ Chapter 13.3 of *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (Friedman, Hastie and Tibshirani).
- ▶ Linear Discriminant Analysis:
 - ▶ Sebastian Raschka's PCA vs LDA article with Python Examples
 - ▶ Chapter 4.3 of *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (Friedman, Hastie and Tibshirani).
- ▶ SVMs:
 - ▶ Jason Weston's SVMs tutorial
 - ▶ e1071 Package for SVMs in R
 - ▶ Chapter 12 of *The Elements of Statistical Learning: Data*